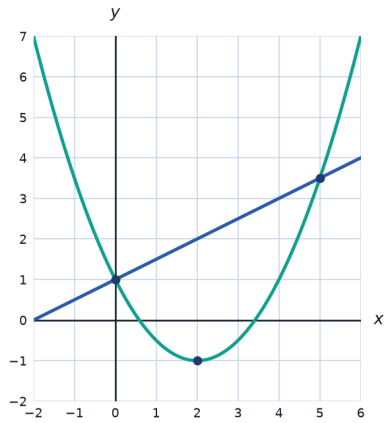
Remainder and factor theorems

Core Concepts: Remainder and factor theorems

1 For the model $f(x) = 6x + 6$ in remainder and factor theorems, evaluate $f(2)$.



a. 18

b. 24

c. 12

d. 22

2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for remainder and factor theorems consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 6x + 6$ in remainder and factor theorems, evaluate $f(3)$.

a. 24

b. 30

c. 18

d. 28

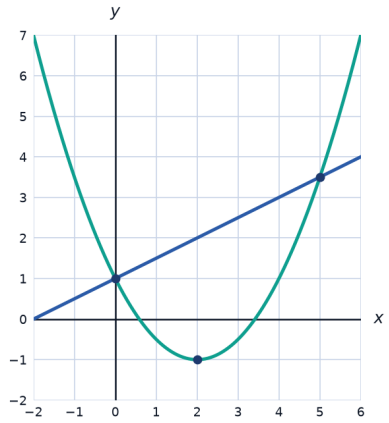
5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$



6 A data set recorded for remainder and factor theorems consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 6x + 6$ in remainder and factor theorems, evaluate $f(4)$.

a. 30

b. 36

c. 24

d. 34

8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

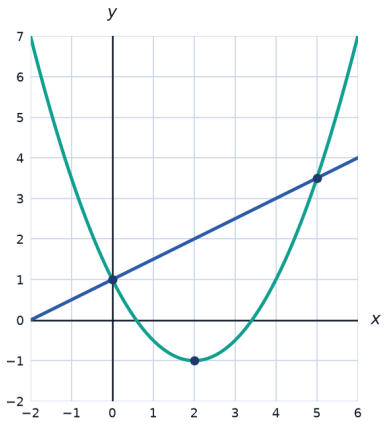
9 A data set recorded for remainder and factor theorems consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

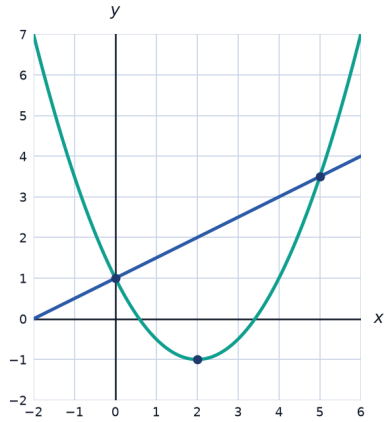
d. 24.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 6x + 6$ in remainder and factor theorems, evaluate $f(5)$.



a. 36

b. 42

c. 30

d. 40

11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

12 A data set recorded for remainder and factor theorems consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

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